

Restaurant-Sales-Performance-SQL-Analytics

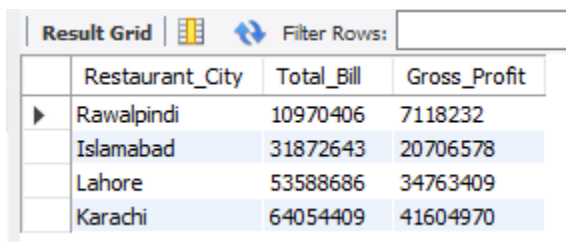
An end-to-end data analysis project exploring a dataset of 100,000+ restaurant transactions using advanced SQL queries. The goal is to uncover actionable business insights regarding revenue distribution, customer payment preferences, city-wise performance, and top-selling menu items to support data-driven decision-making

Business Problems & SQL Solutions

1. City Financial Performance

Business Problem: Calculate total rounded bills and gross profit per city, sorted by lowest total bill first, then highest gross profit.

```
SELECT
    Restaurant_City,
    ROUND(SUM(Total_Bill),0) AS Total_Bill,
    ROUND(SUM(Gross_Profit),0) AS Gross_Profit
FROM rest
GROUP BY Restaurant_City
ORDER BY 2, 3 DESC;
```



The screenshot shows a 'Result Grid' with a 'Filter Rows' button. The grid contains the following data:

	Restaurant_City	Total_Bill	Gross_Profit
▶	Rawalpindi	10970406	7118232
	Islamabad	31872643	20706578
	Lahore	53588686	34763409
	Karachi	64054409	41604970

2. Payment Method Insights

Business Problem: Find the total rounded bill and total order count for each payment method, sorted from highest to lowest total bill.

```
SELECT
    Payment_Method,
    ROUND(SUM(Total_Bill),0) AS Total_Bill,
    COUNT(Order_ID) AS Orders
```

FROM rest

GROUP BY Payment_Method

ORDER BY Total_Bill DESC;

Result Grid				Filter Rows:	
	Payment_Method	Total_Bill	Orders		
▶	HBLPay	23573395	14443		
	JazzCash	23033825	14193		
	Cash	22935555	14269		
	DebitCard	22901459	14273		
	EasyPaisa	22794366	14435		
	BankTransfer	22768717	14231		
	CreditCard	22478826	14156		

3. Top 5 Best-Selling Items

Business Problem: Find the top 5 highest-selling menu items across all categories based on total quantity sold.

SELECT

Category, Item_Name, SUM(Quantity) AS Qun

FROM rest

GROUP BY Category, Item_Name

ORDER BY Qun DESC

LIMIT 5;

Result Grid

Filter Rows:

	Category	Item_Name	Qun
▶	MainCourse	Paya	6551
	MainCourse	ChickenBiryani	6421
	Beverages	FreshLimeSoda	6420
	Sides	GarlicBread	6418
	MainCourse	Haleem	6401

4. Platform Performance

Business Problem: Find the total order count and rounded total bill for each ordering platform, sorted by lowest orders first, then highest total bill.

SELECT

Platform,

COUNT(Order_ID) AS orders,

ROUND(SUM(Total_Bill),0) AS Total_Bill

FROM rest

GROUP BY Platform

ORDER BY orders, Total_Bill DESC;

Result Grid			Filter Rows:	
	Platform	orders	Total_Bill	
▶	Drive-ThroughApp	2522	3961235	
	PhoneOrder	4910	7875922	
	Walk-In	9939	16490290	
	CareemFood	11765	18771978	
	UberEats	11945	19138266	
	DirectCall	12103	19335395	
	FoodPanda	12181	19662259	
	WhatsAppOrder	17023	27302925	
	In-Person	17612	27947874	

5. Revenue Share by City

Business Problem: Calculate each city's percentage share of the overall total revenue.

SELECT

Restaurant_City,

ROUND(SUM(Total_Bill) / (SELECT SUM(Total_Bill) FROM rest) * 100, 1) AS revenue

FROM rest

GROUP BY 1;

Result Grid	Filter Rows:
Restaurant_City	revenue
Islamabad	19.9
Karachi	39.9
Lahore	33.4
Rawalpindi	6.8

6. Top Item per City (Window Functions)

Business Problem: Find the highest-selling menu item (top-ranked by quantity) for each restaurant city.

```
SELECT Restaurant_City, Item_Name, ranks, quan
FROM (
    SELECT Restaurant_City, Item_Name,
    SUM(Quantity) AS quan,
    RANK() OVER (PARTITION BY Restaurant_City ORDER BY SUM(Quantity) DESC) AS ranks
    FROM rest
    GROUP BY 1, 2
) AS a
WHERE ranks = 1;
```

Result Grid

Filter Rows:

Export

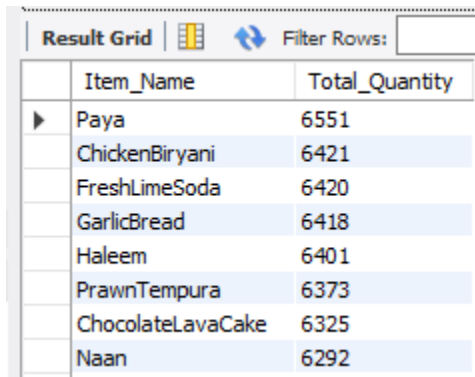
	Restaurant_City	Item_Name	ranks	quan
▶	Islamabad	Paya	1	1395
	Karachi	ChocolateLavaCake	1	2635
	Lahore	Naan	1	2224
	Rawalpindi	ChickenTikka	1	547

7. Above-Average Selling Items (Subqueries & HAVING)

Business Problem: Find all menu items whose total quantity sold is greater than the overall average quantity sold across all items.

```
SELECT Item_Name, SUM(Quantity) AS Total_Quantity
FROM rest
GROUP BY Item_Name
HAVING SUM(Quantity) > (
    SELECT AVG(item_total)
    FROM (
        SELECT SUM(Quantity) AS item_total
```

```
FROM rest
GROUP BY Item_Name
) AS sub
)
ORDER BY Total_Quantity DESC;
```



	Item_Name	Total_Quantity
▶	Paya	6551
	ChickenBiryani	6421
	FreshLimeSoda	6420
	GarlicBread	6418
	Haleem	6401
	PrawnTempura	6373
	ChocolateLavaCake	6325
	Naan	6292

Language: SQL